



Washington University in St. Louis

JAMES MCKELVEY SCHOOL OF ENGINEERING

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Flores Research Group: Additive Manufacturing of Bulk Metallic Glasses

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ABSTRACT: *Bulk metallic glasses show a lot of promise in use as coatings and materials due to their improved strength and corrosion resistance properties. Manufacturing methods, such as additive manufacturing show much promise in increasing the efficiency and success levels of glass formation in addition to the shapes that can be made. Laser Engineered Net Shaping is one option that allows the user to make walls or shapes out of metal powder. Optimizing the parameters to minimize heat input is very important and a very tricky operation that still requires much investigation. The Vit-106a deposited in this experiment did not end up forming glass and instead exhibited unusual wall forming behavior.*

INTRODUCTION

Bulk metallic glasses (BMGs) lack crystalline structure. They are amorphous in structure [1]. Essentially, there is not repeating pattern seen in the microstructure of the material. This is different from other metals which have a repeating pattern throughout. This difference causes a much higher strength, elasticity, resistance to wear and corrosion, and increased hardness [1]. This is largely due to structural defects like grain boundaries and dislocations not having any presence in the BMGs. Currently, a large roadblock in the process of forming metallic glasses is the incredibly high cooling rate required. For a binary alloy system, a fully amorphous structure is achieved with a cooling rate having a magnitude around $10^5 - 10^6$ K/s [2]. However, we are not working with a binary system, so the required cooling rates are lower. The proposed solutions are changing the composition of the alloys to hopefully increase the chances of forming glass, or changing the method used to try and form these glasses. One of these methods is Laser Engineered Net Shaping (LENSTM). This method offers a new way to try and form BMGs so that their properties may be taken advantage of. The LENS is a direct

laser deposition technique [3] and is able to melt small amounts at a high cooling rate. Additionally, it moves in three directions, functioning similarly to a 3D printer. The laser head is stationary on all axes except for the Z-direction. Four powder hoppers have lines to the laser head and are to a desired RPM. The RPM affects how much powder is released into the melt pool. Too low of an RPM means the wall or shape will not build up as tall as desired. On the other side, too high of an RPM, and thus too much powder, will leave a lot of unmelted material. The laser travels forming a melt pool on the substrate. The powder is then released into the melt pool and also melts. Ideally, the size of the melt pool is small enough that we achieve a fast enough quench rate to form a metallic glass.

The LENS allows variation of many parameters, including travel speed, powder feed rate, laser power, layer height, and pathway of the laser head and powder stream. The three most important have a direct impact on the heat input, which provides a simplified optimization path. Equation 1 details the heat input calculation:

$$\lambda = \frac{P}{vD} \quad (1)$$

where P is the laser power, v is the laser travel speed, and D is the laser beam diameter on the substrate [3]. Finding the appropriate heat input is likely key to consistently forming BMGs. Other compositions, such as $Cu_{47}Ti_{33}Zr_{11}Ni_8Si_1$ show better glass forming ability around 9 J/mm^2 [3].

This research focuses primarily on Vit-106a which is composed of 70% Zirconium, 13% Copper, 9% Nickel, 3.65% Aluminum, and 3.4% Niobium [4]. Thermal properties include a melting point of $837 \text{ }^\circ\text{C}$ [4], a crystallization temperature of $499 \text{ }^\circ\text{C}$ [4], and a glass transition temperature of $395 \text{ }^\circ\text{C}$ [4]. Additionally, the powder particles were sieved and had a diameter of approximately $60\text{-}100 \text{ }\mu\text{m}$.

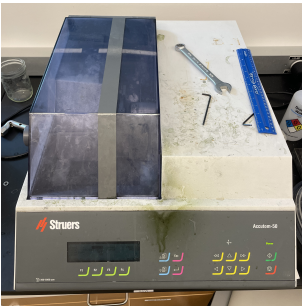
METHODS

Apparatus. Table 1 below depicts the equipment required to prepare, deposit, and analyze.

Table 1 Primary equipment used in the experiment.

Equipment	Quantity	Description	Model Number	Serial Number
Struers Accuton-50	1	Saw used to cut down substrates and cut cross-sections for walls.		
Branson 2510 Sonicator	1	Used to clean dirt and debris off of substrates prior to polishing.		
MoPao 160E Manual Polisher	1	Grind substrate surface to rough it up enough for proper deposition.	812-100	812-100-00003
OPTOMECH MR-7 LENS	1	Laser-deposition for additive manufacturing in an argon environment.		
Nikon Eclipse LV150N	1	Used to examine cross-sections of walls. Optical microscope.	2212-01-11	129-2624
Allied TechPrep Multiprep Polisher	1	Used to polish cross-sections for microscope.		

Figures 1a-1f depict the equipment used to carry out the experiment and collect data.



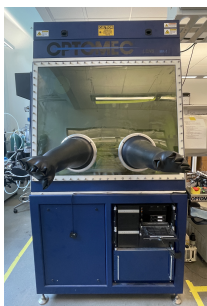
(a) Struers Accuton-50.



(b) Branson 2510 Sonicator.



(c) MoPao 160E.



(d) OPTOMECH LENS.



(e) Nikon Eclipse LV150N.



(f) Allied TechPrep Polisher.

Figure 1 Equipment used in experiment.

Procedure. Two things must happen prior to deposition. First, feedrate calibration must take place to ensure the desired amount of powder is released during deposition. Second, a substrate must be prepared.

To calibrate the powder feedrate: bags are weighed and then placed into the antechamber. The auto-vac cycle is performed: the antechamber empties out all of the air and fills with argon. The bags can then be brought into the main chamber. The calibration can be performed. The hopper RPMS are changed in pre-determined increments and each feed-rate is timed and run for around a minute. During the one minute run-time, the openings of the bags are held around the laser head and as tight of a seal as possible is made. After the bags are collected, the argon is squeezed out prior to sealing by pressing them up against the wall of the OPTOMECH and slowly moving upwards. This is done to not lose any powder. The bags can then be sealed up and put back in the antechamber. Once the inner door is sealed, the bags may be taken out of the antechamber and weighed. The appropriate RPM for the desired gram per minute output can be selected based on calculations normalized to time.

Table 2 below depicts the hopper RPMs tested to determine the appropriate feed-rate correlation for deposition.

Table 2 Feed-Rate calibration matrix.

Powder Feedrate Calibration 04-01-24			
RPM	Deposition Time (s)	Feedrate (g/min)	Notes
10	60	2.137	BPM - 160
13	60	2.266	
16	45	2.44	
19	45	3.04	
22	45	3.333333333	

Next, a substrate was prepared. A Zirconium plate is cut to 1.25" x 0.75" using the Struers Accutom-50. Next, the substrate is sonicated in methanol for 15 minutes to remove debris or dirt. Using the MoPao 160E manual polisher, the substrate is polished for 5 minutes, or until the scratches from the previous direction disappear, at 60 grit, 180 grit, and 300 grit. When increasing the grit of the polishing pad, the substrate is rotated 90 degrees. Once the substrate is ready for deposition, it is taken to the OPTOMECH LENS and loaded into the antechamber.

For deposition, once the sample is in the main chamber the substrate is positioned and secured using washers and bolts. The laser head is lowered until it is just barely touching the laser focusing tool. Next, the laser is set up according to the instruction manual.

Next, the DMC code generated from [A](#) is loaded via flash-drive. The appropriate laser power and powder feedrate are selected in the software and the appropriate hopper, in this case the one containing Vit-106a, is turned on. The execute file button is selected with the mouse hovering over the "abort" button in case of emergency. In order to ensure consistent and smooth flow of powder, the hopper was mechanically agitated by tapping it at 160 BPM. After the deposit, the laser head is brought back to "zero" and moved over to the planned next wall position. This is repeated for each wall deposit and the steps are then reversed to turn off the OPTOMECH safely.

All depositions except for one contain one "pass." A pass is just one line. The deposition goes N-S-N-S (back and forth) with the powder flowing continuously. Two passes essentially overlaps a singular pass with another single pass.

Table [3](#) below details the various parameters for which depositions were created.

Table 3 Parameter test matrix for deposition.

Parameter:	Dep. 1	Dep. 2	Dep. 3	Dep. 4	Dep. 5	Dep. 6	Dep. 7	Dep. 8	Dep. 9	Dep. 10	Dep. 11	Dep. 12
Laser Power (W)	150	150	175	200	225	250	150	150	150	150	150	150
Scanning Speed (mm/s)	21.2	8	8	8	8	8	10	12	14	16	18	8
# of Passes	1	1	1	1	1	1	1	1	1	1	1	2
Heat Input (J/mm ²)	7.0755	18.75	21.875	25	28.125	31.25	15	12.5	10.714	9.375	8.333	18.75

Additionally, the following parameters seen in Table 4 were held constant for each deposit.

Table 4 Parameter test matrix for deposition.

Parameter:	Constant
Wall Height (mm)	6.35
Feedrate (g/min)	3
Hopper RPMs	19
Layer Height (mm)	0.198
Beam Width (mm)	1
# of Layers	32

After the deposition, each row of walls is cross-sectioned perpendicular to the middle of each wall. Initially, an RPM of 5000 and an speed of 0.005 mm/s was used, however, it was determined that an RPM of 1500 was much more effective and thus used for the second large cut.

Next, the newly exposed wall section is polished at 400 grit in 15 minute increments. Each polishing pad is using 2-3 times depending on wear. 5 polishing pads were used. The primary purpose of the polishing was to remove material until the wall had a clearer cross-section than the saw provided. Once enough was removed, the polishing was still done at 400 grit in order to examine under a microscope.

Analysis. The deposits are analyzed by their heat input and visually. To gain a rough estimate of the heat input to the walls we use Eq. 1. Visually, we inspect each wall to determine how "glassy" the sample looks, how much powder was captured and melted, as well as how much powder did not melt.

RESULTS & DISCUSSION

Overall, the walls turned out very interestingly. As can be seen in Fig. 2, they are very uneven, with much more deposition at either end and a sort of divot in the middle.

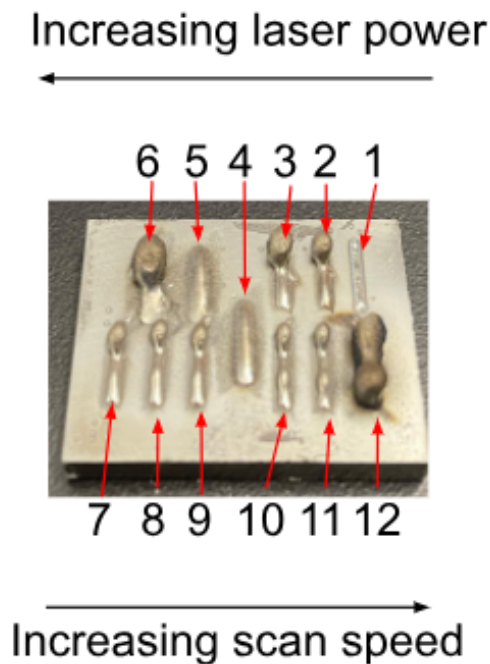


Figure 2 Labeled deposition of Vit-106a results. Each wall is 7mm and the substrate is 1.25"x0.75"

Across walls 1-6 the laser power was increased in increments of 25W with wall 6 containing the highest power. Across walls 7-11, scanning speed was increased in increments of 2, with wall 11 having the highest scanning speed. Wall 12 has the same conditions as wall 2, except two passes are

done instead of one. Wall 1 had an extremely high scan speed which led to virtually no height at all. Walls 4 and 5 look better, but the laser head was 0.25 inches above the laser focusing tool. This could indicate an adjustment of the tool is required.

The cross-sections were done in the middle of each wall section, splitting up the substrate into three separate pieces. This can be seen in Fig. 3, below.



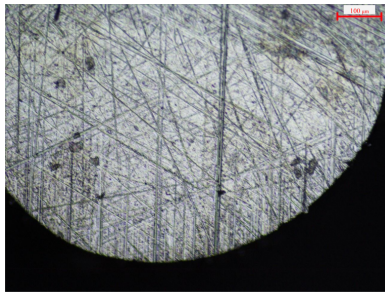
Figure 3 Cross-sectioned walls.

The bottom piece was selected for polishing and analysis as the walls turned out the best, and there was less polishing required to get into the "peaks" at the end of each wall.

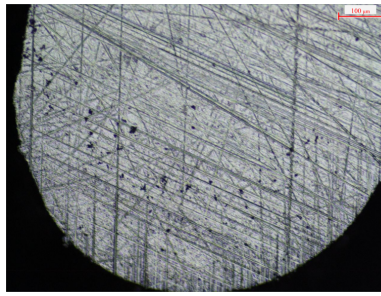
Next, the grain structure and layer variation was examined using optical microscopy. Tables 3-4 can be referred to for more detailed deposition conditions. We will look at depositions 7-12.

The most focus will be given to deposition 12. This wall was made with two passes, which in turn built it up pretty high. Thus, it is the best candidate to look at correlation between grain structure and height, as well as counting the number of visible "layers," checking if they match up with the

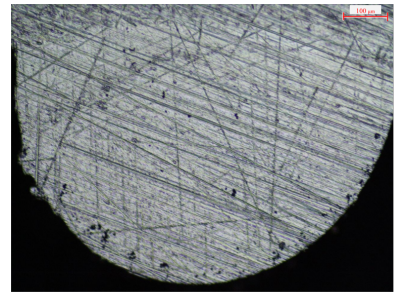
desired number of layers input into the code. Figures 4a-4f depict the various micro-structures of the deposits. Deposits 7-12 were all done at 150W laser power.



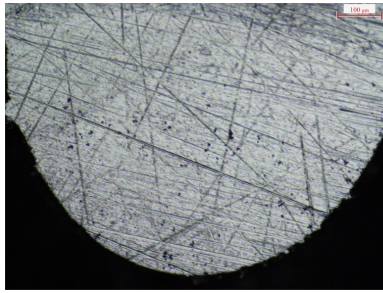
(a) Dep. 7. 10mm/s



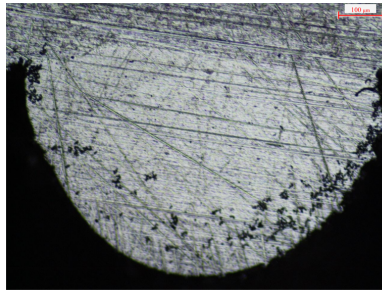
(b) Dep. 8. 12mm/s



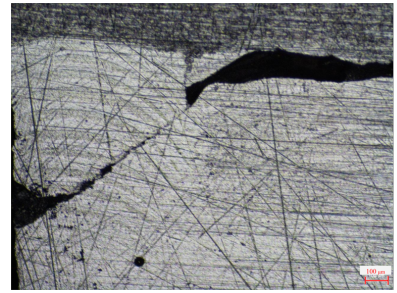
(c) Dep. 9. 14mm/s



(d) Dep. 10. 16 mm/s



(e) Dep. 11. 18 mm/s

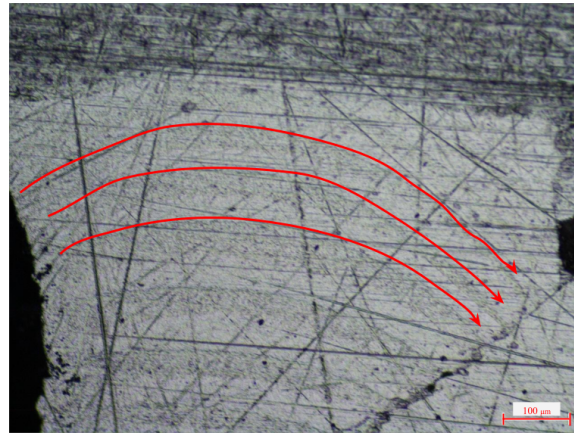


(f) Dep. 12. 8mm/s. 2-pass.

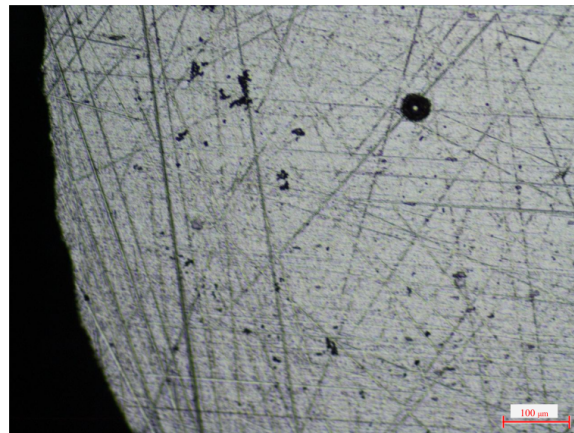
Figure 4 Broad view of each wall deposition cross-section.

Due to the rough nature of the polishing, these specific images do not provide any information on whether the deposition is crystalline or amorphous. Further polishing can be done to determine the nature of the deposits. Furthermore, only the two pass wall has any semblance of layering. The other walls do not appear to show any signs that the powder was put on in layers. Deposit 11 has these little black dots that look like unmelted powder. These are pores in the deposit. Pores are incompletely melted and fused. It makes sense that deposit 11 has the most, as its scanning speed was the highest of all of these deposits. This gave very minimal heat input, which decreased the heat input and increased the cooling rate, but was too fast to fully melt all of the powder.

We then take a closer look at deposit 12 to see if height in the wall affects the grain structure or anything else. Fig. 5 depicts the cross-section of deposit 12.



(a) Dep. 12 lower



(b) Dep. 12 higher

Figure 5 Cross-section of deposit 12. 150W, 8mm/s, 2-pass.

In Fig. 5a we see that there appear to be layers swooping upwards from the outside to the inside. Oddly, this is not seen in any of the other deposits. However, looking at Fig. 5b, these lines don't continue up throughout the wall. Seeing as the walls are made in layers and each one had 32 passes, this is an interesting result. It could simply be an indication that the powder mixed well. However, it could also be an indication that the higher we get in the wall, or the more central we are

in the wall, the more heat that is conducting through the whole entire wall, melting the layers together more than anticipated.

The desired height of each wall was 6.35 mm, which is about 0.25". The wall heights are measured from the cross sections, which are located at the middle of each wall. The wall heights compared to scanning speed are seen below in Table 5.

Table 5 Heights of Depositions 7-12

Parameter:	Dep. 7	Dep. 8	Dep. 9	Dep. 10	Dep. 11	Dep. 12
Scanning Speed (mm/s)	10	12	14	16	18	8
Heat Input (J/mm ²)	15	12.5	10.714	9.375	8.33	18.75
Wall Height (mm)	0.58	0.65	0.59	0.6	0.5	2.8
Difference From Desired Wall Height (mm)	5.77	5.7	5.76	5.75	5.85	3.55

Note that the ends of wall 12 appear to be nearly double in height, which gets us very close to the desired 6.35 mm. Although the deposit has the highest heat input, the wall is taller because of the second pass. Due to the slight overlap, the edges of the walls are rounded, but the center overlapping section can grow very tall. Otherwise, when held at 150W, scanning speed between 10-18 mm/s doesn't seem to have too much of an impact on the height of the wall. It seems as though laser power has more of an impact on the height of the wall and scan speed can be selected to ensure too much powder isn't deposit.

We see lots of scratches throughout all of the deposits, this is due to only polishing up to 400 grit.

There are a few possible sources of error present in this experiment. First, as can be seen in Fig.

2, the walls are not uniform. This is likely due to the acceleration and deceleration at the beginning and end of each wall, when the platform has to slow down, stop, switch direction, and increase speed again. Additionally, there are issues with the Vit-106a powder feeding through the hopper. It is not very consistent and does not flow well unless the hopper is being shaken up by force from a wrench. Part of this is due to the age of the powder and the difficulty in sourcing new powder with the right specifications and manufacturing processes.

CONCLUSION

The formation of BMGs in wall shapes is a very complex problem that still needs a lot of research on different parameters to create a "formula" that works every time. An RPM of 19 is too high and a lower value should be chosen and experimented on. Previous work was done at 8 RPMs, so ideally a midpoint value would be selected to gather data from there. Additionally, a two pass wall does the best at building overall wall height, but there isn't a ton of variation in crystal structure throughout the wall itself. The two pass likely added so much wall height because double the powder was being added for each layer. Additionally, the slower scan speed likely allowed for more powder to melt.

Ultimately, it seems unlikely that the parameters tested in this research did lead to the formation of glass. The next iteration would likely focus in on a middle ground RPM and creating consistent walls.

Acknowledgements

Special thanks to Professor Flores, Abner Peralta, and Jackson Ford.

References

- [1] Sohrabi, N., Jhabvala, J., and Loge, R., 2021, “Additive Manufacturing of Bulk Metallic Glasses-Process, Challenges and Properties: A Review,” *Metals*, **11**.
- [2] Zhang, C., Ouyang, D., Pauly, S., and Liu, L., 2021, “3D Printing of Bulk Metallic Glasses,” *Materials Science and Engineering R*.
- [3] Sun, H. and Flores, K. M., 2008, “Laser Deposition of a Cu-based metallic glass powder on a Zr-based glass substrate,” *J. Mater. Res.*, **23**(10), pp. 2692–2703.
- [4] LLC, M., “MatWeb: Online Materials Information Resource: Materion Vit 106a Zirconium-Based Bulk Metallic Glass (BMG),” <https://www.matweb.com/search/datasheet.aspx?matguid=5c9ee48e2a704449b5880173789349cb&ckck=1>

A MATLAB Code

The following code was created by Jackson Ford.

```
1  clc
2  close
3  clear
4
5  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% READ ME %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
6  % The following program takes in a rectangular prism shaped wall
7  % (dimensions specified by user) and converts it to DMC code to be read
8  % by the Optomec LENS MR-7 located in Flores Research Group Laboratory.
9
10 % The code is designed to take in a list of parameters (scan speed (mm/s),
11 % layer height (mm), number of passes, and scan pattern (WIP); however,
12 % laser power (W), powder feedrate (g/min), and beam width (mm) are
13 % modified outside of the DMC code, on the operating computer itself) that
14 % are being tested for metallic glass production.
15
16 % This code utilizes a convention where a DMC vector move is utilized when
17 % the laser is on, and DMC coordinate moves when the laser is off.
18
19 % NOTE: Vector moves are incremental, while coordinate moves are absolute
20
21 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
22
23 % CREATE DMC FILE
24 prompt = 'What would you like to name your DMC file?\n';    % input ?
```

```

25 filename = input(prompt, 's');    % Ask user for file name
26 filename = filename + ".dmc";    % Create filename into DMC file
27
28 if isfile(filename)              % If file exists, overwrite it.
29     delete(filename)
30     diary(filename)
31 else                             % If it doesn't, create it.
32     diary(filename)
33 end
34
35 % DEBUG VARIABLES
36 debugvar = 0;
37
38 % USER DEFINED VARIABLES
39 % Processing Parameters
40 scanSpd = 8;                    % defines the scanning speed in mm/s
41 numPasses = 2;                  % defines # of passes per layer
42 numLayers = 32;                 % defines # of layers; based on layer and wall height
43 hatchorcrosshatch = 0;         % IN DEVELOPMENT (could be used for add. param.)
44
45 % Wall Characteristics
46 lengthOfWall = 7;               % defines wall length in mm
47 widthOfWall = 1;                % defines wall width in mm
48 heightOfWall = 6.35;            % defines wall height in mm
49
50 % Machine Parameters
51 resolution = 200000;             % resolution of LENS (typ. 200000)
52 convert = resolution * (1/25.4); % used to convert mm to counts in dmc
53 VA = 3000000;                   % defines incremental acceleration, in cts/s^2

```

```

54 VD = 3000000;           % defines incremental deceleration, in cts/s^2
55 VS = round(scanSpd * convert);    % inc. speed calc. from scanning speed
56 AC = 3000000;           % defines absolute acceleration, in counts/s^2
57 DC = 3000000;           % defines absolute deceleration, in counts/s^2
58 SP = VS;                % defines absolute speed, in counts/s
59
60 % COMPUTER CALCULATED VARIABLES
61     % Convert mmm wall to counts wall
62 countWidth = widthOfWall * convert;
63 countLength = lengthOfWall * convert;
64 countHeight = heightOfWall * convert;
65
66     % Create DMC motion text variables
67 VAtext = ['VA ', num2str(VA)];
68 VDtext = ['VD ', num2str(VD)];
69 VStext = ['VS ', num2str(VS)];
70 ACtext = ['AC ', num2str(AC), ', ', num2str(AC), ', ', num2str(AC)];
71 DCtext = ['DC ', num2str(DC), ', ', num2str(DC), ', ', num2str(DC)];
72 SPtext = ['SP ', num2str(SP), ', ', num2str(SP), ', ', num2str(SP)];
73
74 % DECLARATION OF COORDINATE/VECTOR VARIABLES
75 originx = 0;
76 originy = 0;
77 xcoord = 0;
78 ycoord = 0;
79 zcoord = 0;
80 oldxcoord = 0;
81 oldycoord = 0;
82 oldzcoord = 0;

```

```

83 xvector = 0;
84 yvector = 0;
85 zvector = 0;
86
87 % BEGINNING PORTION OF DMC CODE
88     % Slicer Comments that are placed at the beginning of all our DMC files
89 disp('REM SLICE1 PWR  35.0 PF1=1');
90 disp('REM SLICE2 PWR  35.0 PF2=2');
91 disp('REM SLICE PWR   END');
92 disp(['REM *****' ...
93     '*****']);
94 disp(['REM C:\Documents and Settings\Administrator\Desktop\HEA-5x5-' ...
95     'Square_30_y_v2.0.DMC']);
96 disp(['REM *****' ...
97     '*****']);
98 disp('REM ***** Convert Slice to DMC Process Parameters *****');
99 disp('REM Layer Thickness = 0.008 ');
100 disp('REM Resolution = 5000 ');
101 disp('REM Contour Feedrate = 40 ');
102 disp('REM X Axis Resolution = 200000 ');
103 disp('REM Y Axis Resolution = 200000 ');
104 disp('REM Z Axis Resolution = 200000 ');
105 disp('REM Laser On Feedrate = 40 ');
106 disp('REM Laser On Accel = 60000 ');
107 disp('REM Laser On Decel = 60000 ');
108 disp('REM Laser On Shutter Delay = 20 ');
109 disp('REM Laser Off Feedrate = 60 ');
110 disp('REM Laser Off Accel = 60000 ');
111 disp('REM Laser Off Decel = 60000 ');

```



```

112 disp('REM Laser Off Shutter Delay = 20      ');
113 disp('REM *****');
114
115     % Begin writing LENS readable code
116 disp('DP 0,0,0'); % Sets current position as origin
117 disp(ACtext); % Declares abs. accelerate in DMC
118 disp(DCtext); % Declares abs. accelerate in DMC
119 disp(SPtext); % Declares abs. accelerate in DMC
120 disp('WT 20000'); % Wait 20 seconds
121
122 % IF CROSSHATCH IS ENABLED (Still under development)
123 if hatchorcrosshatch == 0
124     for layers = 1 : numLayers % Repeats for each layer
125         % Calculate Height
126         if layers == 1
127             zHeight = 0;
128         else
129             zHeight = (layers - 1) * (countHeight / (numLayers - 1));
130         end
131
132         % Plot layer change for even layers
133         if layers > 1 && mod(numPasses, 2) ≠ 0
134             quiver3(xcoord / convert, ...
135                     yvector / convert, ...
136                     prevZHeight / convert, ...
137                     -xcoord / convert, ...
138                     -yvector / convert, ...
139                     (countHeight / (numLayers - 1)) / convert, ...
140                     'b-', 'LineWidth', 1, 'MaxHeadSize', 50)

```

```

141     end
142
143     % Plot layer change for even layers
144     if layers > 1 && mod(numPasses, 2) == 0
145         quiver3(xcoord / convert, ...
146                 0, ...
147                 prevZHeight / convert, ...
148                 -xcoord / convert, ...
149                 0, ...
150                 (countHeight / (numLayers - 1)) / convert, ...
151                 'b-', 'LineWidth', 1, 'MaxHeadSize', 50)
152     end
153
154     % Write DMC Code
155     xcoord = originx;
156     ycoord = originy;
157     zcoord = zHeight;
158     oldxcoord = xcoord;
159     oldycoord = ycoord;
160     oldzcoord = zcoord;
161     xvector = 0;
162     yvector = 0;
163     zvector = zHeight;
164     PAtextdisplay = PAtext_calculate(xcoord, ycoord, zcoord, ...
165                                     oldxcoord, oldycoord, ...
166                                     xvector, yvector, zvector, ...
167                                     hatchorcrosshatch, convert);
168     disp(PAtextdisplay);    % Sets move coordinate
169     disp('BG XYZ');        % Moves print head

```

```

170     disp('AM XYZ');           % Waits for print head to finish movement
171     passes = 1;
172
173     while passes ≤ numPasses
174         xvector = 0;
175         yvector = countLength;
176         zvector = zHeight;
177         VPtextdisplay = VPtext_calculate(xvector, yvector, zvector, ...
178                                           xcoord, ycoord, zcoord, ...
179                                           hatchorcrosshatch, convert);
180         disp('VM XY');         % Prepare for vector move
181         disp(VAtext);          % Sets inc acceleration
182         disp(VDtext);          % Sets inc deceleration
183         disp(VStext);          % Sets inc speed
184         disp(VPtextdisplay);   % Declares inc movement
185         disp('VE');            % End vector preparation
186         disp('SB 1');          % Open laser shutter
187         disp('WT 20');         % Wait for shutter to open
188         disp('BGS');           % Complete vector move
189         disp('AMS');           % Wait for movement to complete
190         disp('CB 1');          % Close laser shutter
191         disp('WT 30');         % Wait for shutter to close
192         passes = passes + 1;
193
194         % Breaks the loop if you have just finished your last pass
195         if passes > numPasses
196             debugvar = debugvar+1;
197             prevZHeight = zHeight;
198             break

```

```

199     end
200
201     oldxcoord = xcoord;
202     oldycoord = ycoord;
203     xcoord = xcoord + (countWidth/(numPasses - 1));
204     ycoord = ycoord + countLength;
205     zcoord = zHeight;
206     PAtextdisplay = PAtext_calculate(xcoord, ycoord, zcoord, ...
207                                     oldxcoord, oldycoord, ...
208                                     xvector, yvector, zvector, ...
209                                     hatchorcrosshatch, convert);
210     disp(PAtextdisplay);      % Sets move coordinate
211     disp('BG XYZ');          % Moves print head
212     disp('AM XYZ');          % Waits for print head to finish move
213
214     xvector = 0;
215     yvector = -countLength;
216     zvector = zHeight;
217     VPtextdisplay = VPtext_calculate(xvector, yvector, zvector, ...
218                                     xcoord, ycoord, zcoord, ...
219                                     hatchorcrosshatch, convert);
220     disp('VM XY');           % Prepare for vector move
221     disp(VAtext);            % Set vector acceleration
222     disp(VDtext);            % Set vector deceleration
223     disp(VStext);            % Set vector speed
224     disp(VPtextdisplay);     % Declare movement coordinates
225     disp('VE');              % End vector preparation
226     disp('SB 1');            % Open laser shutter
227     disp('WT 20');           % Wait for shutter to open

```

```

228         disp('BGS');           % Begin movement
229         disp('AMS');           % Wait for movement to finish
230         disp('CB 1');          % Close laser shutter
231         disp('WT 30');         % Wait for shutter to close
232         passes = passes + 1;
233
234         % Breaks the loop if you have just finished your last pass
235         if passes > numPasses
236             debugvar = debugvar+1;
237             prevZHeight = zHeight;
238             break
239         end
240
241         oldxcoord = xcoord;
242         oldycoord = ycoord;
243         xcoord = xcoord + (countWidth/(numPasses - 1));
244         ycoord = ycoord - countLength;
245         zcoord = zHeight;
246         PAtextdisplay = PAtext_calculate(xcoord, ycoord, zcoord, ...
247             oldxcoord, oldycoord, ...
248             xvector, yvector, zvector, ...
249             hatchorcrosshatch, convert);
250         disp(PAtextdisplay);     % Set move coordinates
251         disp('BG XYZ');         % Moves print head
252         disp('AM XYZ');         % Waits for print head to finish move
253     end
254 end
255 end
256

```

```

257 disp("UI 3");      % End DMC program

258

259 diary off % Stops writing dmc file

260

261 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
262 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% INNER SCRIPT FUNCTIONS BELOW %%%%%%%%%
263 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
264
265
266 % Function to create coordinate moves into DMC text and plot on slicer
267 function p = PAtext_calculate(xcoord, ycoord, zcoord, ...
268                               oldxcoord, oldycoord, ...
269                               xvector, yvector, zvector, ...
270                               hatchorcrosshatch, convert)
271
272 % This creates the text for the absolute movement
273 PAtext = ['PA ', ...
274           num2str(round(xcoord)), ',', ...
275           num2str(round(ycoord)), ',', ...
276           num2str(round(zcoord))];
277
278 p = PAtext; % Output variable so it can be used in script
279
280 % This plots what the absolute movement is
281 hold on
282 plot3(xcoord / convert, ...
283        ycoord / convert, ...
284        zcoord / convert, ...
285        'bo', LineWidth=2);

```

```

286     xlabel('X-Axis (mm)');
287     ylabel('Y-Axis (mm)');
288     zlabel('Z-Axis (mm)');
289     title('Slice Image of Wall');
290     axis equal
291
292     if hatchorcrosshatch == 0
293         % Plots absolute move on slice image
294         quiver3((oldxcoord + xvector) / convert, ...
295                (oldycoord + yvector) / convert, ...
296                zvector / convert, ...
297                (xcoord - oldxcoord - xvector) / convert, ...
298                (ycoord - oldycoord - yvector) / convert, ...
299                0, ...
300                'b-', 'LineWidth', 1, 'MaxHeadSize', 50)
301     end
302 end
303
304
305 % Function to create vector moves into DMC text and plot on slicer
306 function v = VPtext_calculate(xvector, yvector, zvector, ...
307                               xcoord, ycoord, zcoord, ...
308                               hatchorcrosshatch, convert)
309
310 % This creates the text for the incremental movement
311 VPtext = ['VP ', ...
312           num2str(round(xvector)), ', ', ...
313           num2str(round(yvector))];
314

```

```

315     v = VPtext;    % Output variable so it can be used in script
316
317     % This plots the point to where the vector movement is going
318     hold on
319     plot3((xcoord + xvector) / convert, ...
320           (ycoord + yvector) / convert, ...
321           zvector / convert, ...
322           'ro',LineWidth=2)
323     axis equal
324
325     if hatchorcrosshatch == 0
326         % This plots the incremental directional arrow
327         quiver3(xcoord / convert, ycoord / convert, zcoord / convert, ...
328                xvector / convert, yvector / convert, 0,...
329                'r-',LineWidth=1,MaxHeadSize=1)
330     end
331 end

```